



AKSHAYKUMAR GUNARI

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CAREER OBJECTIVE

Seeking a dynamic job in a highly regarded company where I can confidently utilize my skills and flair to bring about a productive change within myself and thus aim towards a broader perspective of developing along with the company.

EXPERIENCE

- Juniper Networks India Private Limited** Bangalore, India
Software Engineer - 2 March 2024 - Present 0 year(s) and 10 month(s)
 - Working in the Linux Platform Development Team on bringing up QFX-5K Series Switches.
 - My responsibilities include the development of various daemons (on **Junos EVO OS**: a unified, end-to-end network OS widely deployed in most of the Juniper Networks products) related to the platform side (Routing Engines) on the Networking switches, mainly designed for Data Centers.
- AMD India Private Limited** Bangalore, India
System Software Designer - 2 May 2023 - March 2024 0 year(s) and 10 month(s)
 - I Moved to the Linux Platform Driver Team for Server BU, where I was involved mostly in DMA, Non-Transparent Bridge device driver development activities on Server Platforms.
 - Also worked on the development of linux tools which helped in the performance evaluation of the DMA Engine, **dma-perf** was one such tool which was customized integration of **dma-test** and **ntb_perf** tool.
System Software Designer - 1 July 2021 - May 2023 1 year(s) and 10 month(s)
 - I was part of Sensor Fusion Hub team, where I was responsible for the enablement of various **Human Presence Detection (HPD)** sensors and Ambient Light Sensors (ALS) on the ARM Cortex M4 co-processor.
 - I was responsible for the enablement of firmware for TDK CH201, Altek Sunny M1 HPD sensors on **AMD MP2**.
- Centre of Excellence in Visual Intelligence** Hubli, India
Research Intern March 2021 - July 2021 0 year(s) and 4 month(s)
 - Worked on a DST project under Dr. Uma Mudenagudi's guidance towards categorizing crowdsourced data involving 2D and 3D data.
 - During my tenure at CEVI, I got a chance to publish four research papers at various national and international conferences.
- Indian Institute of Technology - Delhi** New Delhi, India
Project Trainee June 2019 - July 2019 0 year(s) and 2 month(s)
 - Worked on a project - Endoscopic Tool Tracking which was a Collaborative Neuro-Engineering Platform Building for Excellence in Innovation and Translational Research (RP03571G) [Internship Certificate](#)

EDUCATION

- Birla Institute of Technology** Pilani, India
Masters of Technology in Artificial Intelligence and Machine Learning May 2024 - Pursuing
 - GPA: Pursuing
- KLE Technological University** Hubli, India
Bachelor of Engineering in Computer Science August 2017 - July 2021
 - GPA: 9.02/10.00
- Chetan PU Science College** Hubli, India
Pre-University Education (PCMS) May 2017
 - Grade: 90.5%
- Chetan Public School** Hubli, India
Secondary Education March 2015
 - Grade: 94.40%

PROJECTS

- NTB and DMA Driver Development: [Associated with AMD India Pvt Limited]** January 2023 - March 2024
Domains: [Linux Kernel, DMA, Non Transparent Bridge]
 - I have worked as a Linux Kernel Engineer to bring up the Linux kernel driver for Non-Transparent Bridge and DMA Controllers on different Server Platforms.
 - I was also a key contributor in developing various in-house helping kernel modules which was designed to get the performance evaluation of NTB data transfer over DMA between two server platforms.
- Designing and Enablement of Zephyr RTOS: [Associated with AMD India Pvt Limited]** March 2022 - December 2022
Domains: [RTOS, Zephyr, ARM Processors, Device Tree]
 - Designed, developed, and ported the Zephyr RTOS to the ARM Cortex-M4 co-processor whose functionality was mainly around the various sensors clumped together, making the client platforms smarter.
 - Ported changes related to the device tree, memory configurations, and sensor data analysis.
- HPD Sensors Enablement: [Associated with AMD India Pvt Limited]** July 2021 - December 2022
Domains: [Firmware, Sensors Enablement, I2C Protocol]

- To enable various camera-based and non-camera-based **Human Presence Detection (HPD)** Sensors and Color Sensors on ARM Cortex-M4 co-processor. Built an in-house library for enablement of various other sensors like accelerometer (+gyrometer) and magnetometer.
- To channelize the data communication from the firmware through the driver to the service running on the Windows OS.
- **Progressive Clustering: [Associated with Centre of Excellence in Visual Intelligence (CEVI)]** January 2021 - July 2021
Domain: [Deep Learning, Computer Vision, Incremental Learning, Unsupervised Learning] [9]
 - As the Research Experience for Undergraduates (REU), we proposed a framework to find the arbitrarily shaped clusters, incremental in nature, with dynamically growing datasets exhibiting the increment of the dataset not just with respect to class (Class-Incremental) but also in the distribution of each class.
- **ABD-Net: [Associated with Centre of Excellence in Visual Intelligence (CEVI)]** March 2021 - June 2021
Domains: [Deep Learning, Computer Vision, 3D Vision, Point Clouds, Unsupervised Learning] [9]
 - As a research interns group, my team proposed **Attention Based Decomposition - Network (ABD-Net)**, for point cloud decomposition into basic geometric shapes namely, plane, sphere, cone and cylinder.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] Akshaykumar Gunari, et al. (2022). **Progressive Clustering: An Unsupervised Approach Towards Continual Knowledge Acquisition of Incremental Data**. In *ICPRAI 2022 - 3rd International Conference on Pattern Recognition and Artificial Intelligence*. (Conference held in Paris, June 01), pp. 355–367. Publisher. 01 June 2022, Paris. DOI: 10.1007/978-3-031-09282-4_30
- [C.2] Akshaykumar Gunari, et al. (2022). **ABD-Net: Attention Based Decomposition Network for 3D Point Cloud Decomposition**. In *The first Workshop on Structural and Compositional Learning on 3D Data - ICCV 2021*, IEEE Xplore. 16th October 2021, Held Virtually. DOI: 10.1109/ICCVW54120.2021.00232
- [C.3] Akshaykumar Gunari, et al. (2020). **Augmented Data as an Auxiliary Plug-In Toward Categorization of Crowdsourced Heritage Data**. In *Workshop on Digital Heritage (WDH) 2021*, pp. 53–62. Springer, Singapore. 22nd December 2021, IIT-Jodhpur, India. DOI: 10.1007/978-981-19-4136-8_4
- [C.4] Akshaykumar Gunari, et al. (2020). **Deep Visual Attention-Based Transfer Clustering**. In *Fourth International Conference on Computing and Network Communications (CoCoNet'20)*, pp 357–366. Springer, Singapore. 14th October 2020, Chennai, India. DOI: 10.1007/978-981-33-6987-0_29
- [P.1] Siddharth Katageri, Akshaykumar Gunari, Shashidhar V Kudari, et al. (2022). **System for Attention-Based Decomposition of 3D Point Clouds**. Registration Date: 18th March 2022.

SKILLS

- **Programming Languages:** C, C++, Python, Shell
- **Industry Knowledge:** Operating Systems, Linux Kernel Programming, Device Drivers, Machine learning, Yocto Project, RTOS.
- **Tools:** Git, Jupyter Notebook, Spyder, Pycharm, VS Code, Google Colab, Tensorflow, Pytorch.
- **Data Science & Machine Learning:** Deep learning, Image Processing, Computer Vision

HONORS AND AWARDS

- **Spotlight Award** December 2021
AMD India Private Limited
 - Developed a feature for measuring the various Firmware Boot timestamps during an S2-Idle Cycle in **Smart Trace Buffer (STB)** Parser.
 - It was a crucial requirement from Google for their Chromebook laptops.
- **Spotlight Award** January 2022
AMD India Private Limited
 - As a lab warrior spot light award
- **Spotlight Award** December 2022
AMD India Private Limited
 - For enabling a CH201 - TDK InvenSense HPD Sensor.
 - This was a requirement from HP to integrate five class of sensors (Accelerometer, Gyroscope, Magnetometer, Ambient Light, and HPD Sensors) on the co-processor.

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- **Workshops and Events** August 2018 - July 2021
KLE Technological University [9]
 - Conducted a workshop on the Basics of Image processing, Machine Learning, and volunteered in various cultural and technical events conducted in college.
 - Attended a workshop on 3D Computer Vision (Sponsored) at IIIT-Hyderabad
 - Competed in a Kaggle Competition titled “**Categorical Feature Encoding Challenge**” and ranked **52/1342** - December 9, 2019.
 - Participated in TCS Tech-bytes, Smart India Hackathon - 2019.

CERTIFICATIONS

- **Stanford University on Coursera: Machine Learning Course** August 2020
- **NPTEL Certification: Problem Solving In C** January 2018
- **NPTEL Certification: Data Structures And Algorithms in Python** January 2019

PERSONAL DETAILS

Father's Name: Shrinivas Gunari
Mother's Name: Akshata Gunari
Date of Birth: 18/08/1999
Languages Known: English, Kannada, Hindi.
Interests/Hobbies: Playing Volleyball, Listening to Music, Traveling, and Surfing.
Permanent Address: H. No 137, Akshay Colony, 1st Phase, Gokul Road, Hubli, 580-030.

Declaration: I, hereby declare that the above-provided information is true to the best of my knowledge.

Date: Tuesday 28th January, 2025



(Akshaykumar Gunari)